

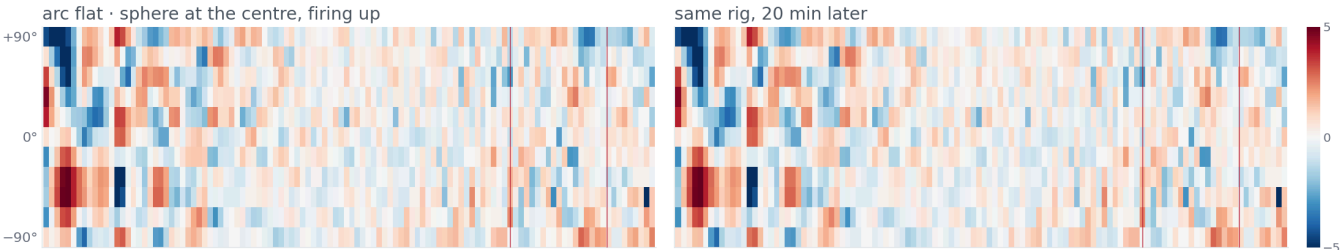
The arc is a good instrument; the map it draws is the room. Below 3 kHz the hub loudspeaker is a valid axisymmetric source, and from 3–5 kHz it is not.

Change one thing at a time

Eleven UMIK-2 capsules on a 1.68 m arc · 112 tones, 257–6350 Hz · each cell is one capsule at one tone, dB re the eleven-capsule mean per-capsule calibration applied · red rules mark 3–5 kHz, where the source is not axisymmetric
the sphere sits on a tripod at the centre of the arc and fires along its axis, so every capsule is the same distance away and at the same angle to it

1 Nothing changed
the same capture twice, 20 min apart

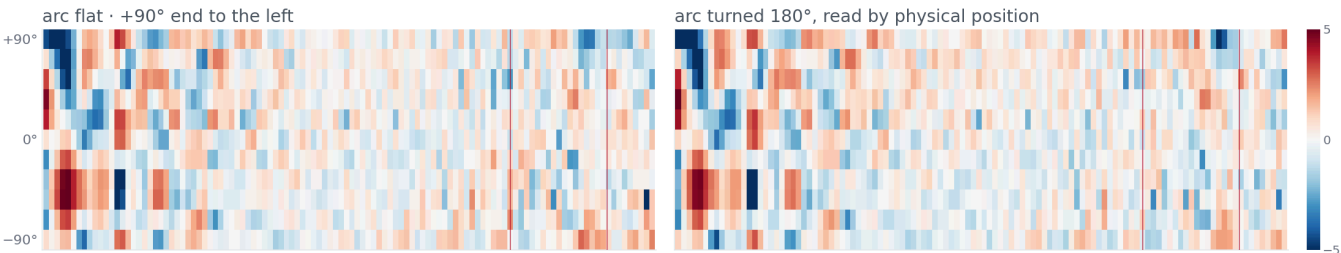
0.03 dB
r = +0.99



the instrument is silent when nothing moves

2 The arc flipped 180°
second run re-indexed by physical position

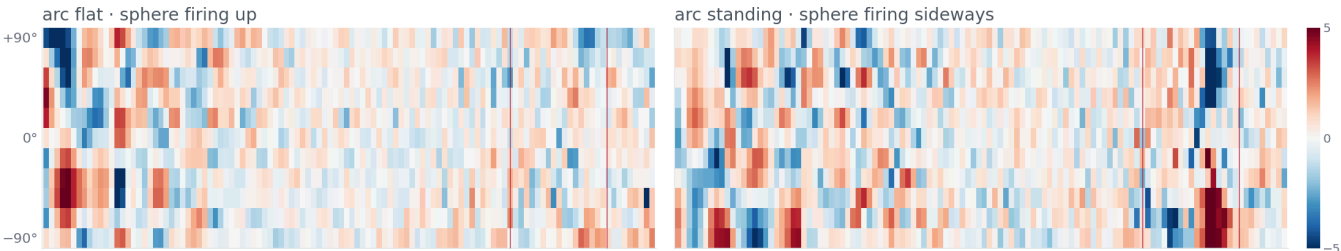
0.53 dB
r = +0.92



the pattern stayed with the room, not with the arc

3 The arc stood vertical
same capsules and cables, different room

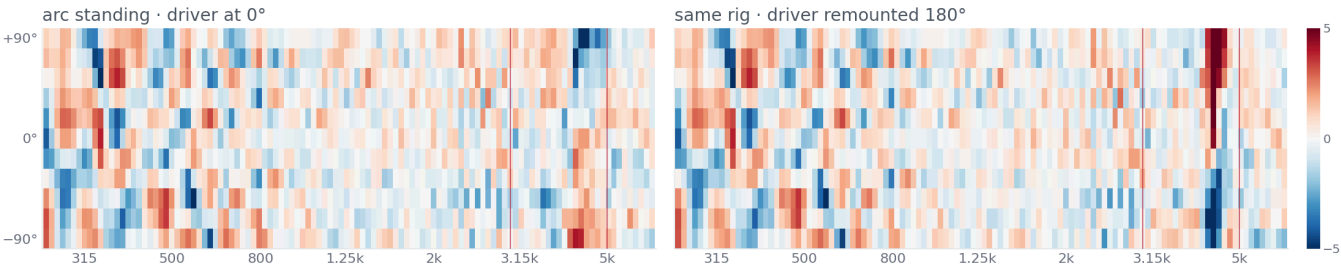
2.29 dB
r = -0.12



change the room and the map is unrecognisable

4 Only the driver turned 180°
in a sphere that never moved

0.20 dB
below 3 kHz



and 11.0 dB at 4362 Hz — the source's own rocking mode

0.21 dB
of the position map is provably the rig
— antisymmetric under the flip

1.04 dB
is provably the room; the 0.84 dB a
single flip cannot divide is room too
(cov = -0.09 dB², CI [-0.22, +0.03])

≤ 0.16 dB
is all the two geometries share — so
the rig's total contribution is ~0.2 dB

11.0 dB
at 4362 Hz when only the driver turns:
an m = 1 rocking mode, ka = 2.39. Avoid
3–5 kHz

Open. The loudspeaker map does not reproduce the propeller map (r = -0.04 tone-notched, +0.04 mixed; all 22 relabellings tested, best +0.24) — the propeller's own inflow, or the object at the hub, still matters. The absolute anchor needs a 94 dB calibrator. Room geometry is unrecorded: hub height, room L×W×H, foam, what stands near the arc ends.
Data. calibrator/sessions/2026-09-16 (capsules) and 2026-09-17 (arc); levels and meta mirrored to the measurement repo, WAVs local. calibrator/rig.py captures and read_map() is the single loader, so a figure cannot disagree with the capture about what a level means; the bad band is SOURCE_SUSPECT_HZ. Every number here was recomputed from those captures while this page was written. Where this page and CLAUDE.md disagree, CLAUDE.md governs. rev 2, 2026-09-17.